Research Report

Predicting Polypharmacy Side Effects with Graph RAG and an LLM

Group 8

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Contents

1 Introduction **1**
1.1 Problem and Motivation 1
1.2 Related Work 1
1.2.1 What is already solid 1
1.2.2 What is still missing 1
1.3 Objective and Project Scope 2
1.3.1 Objective 2
1.3.2 Dataset and suitability 2
1.3.3 Deliverable and boundaries 2
1.4 Report Structure 3

2 Background **3**
2.1 The Knowledge Graph 3
2.2 Standard Graph-RAG 3
2.3 Problem Statement 4

3 Methodology **4**
3.1 Extending Retrieval from Two to $k \geq 3$ Drugs 4
3.2 Pruning Before Passing to the LLM 5
3.3 Output: Tiered, Traceable Predictions 5

Introduction

1.1 Problem and Motivation

Polypharmacy—the concurrent use of several medications by one patient—is common among older adults and patients with chronic or complex diseases. Although often necessary, it raises the risk of drug–drug interactions (DDIs), especially polypharmacy side effects: adverse events that emerge only when two or more drugs are taken together and cannot be attributed to a single drug. Such interactions contribute to hospitalisation, morbidity, and post-market drug withdrawal; identifying harmful combinations early is therefore a direct patient-safety concern.

The problem is fundamentally combinatorial. With thousands of approved drugs, possible combinations are too numerous for exhaustive clinical testing, and many interactions surface only after market release. Computational screening is consequently essential. Drugs, their protein targets, and protein interactions form a natural network, while a side effect between drugs can be represented as a typed edge. This project uses that perspective and the Decagon (BioSNAP) dataset as its biomedical knowledge graph.

1.2 Related Work

1.2.1 What is already solid

Multi-relational link prediction on a drug–protein graph is a well-established and effective formulation. Decagon [1] introduced it, while TIP [2] substantially improved efficiency, reporting AUPRC 0.890 (a 7.2% improvement over Decagon), about 83× faster execution, and roughly threefold lower memory use. Strong results have also been reported by molecular-structure approaches such as SSI-DDI [3] and biomedical-knowledge-graph approaches including KnowDDI [4] and MKG-FENN [5].

Most relevant here, a knowledge graph combined with a large language model (LLM) adds an interpretable explanation layer to prediction. DDI-GPT [6] retrieves a subgraph around a drug pair, converts it to text, and fine-tunes an LLM to return both a prediction and a gene-level rationale. More broadly, large-scale evaluation indicates that even modest LLMs can predict DDIs effectively [7]. Recent systems extend this direction with mechanism-conditioned multi-agent reasoning (DDIAgents [9]) and dynamic multi-view graph-and-language reasoning (DrugReason [10]).

1.2.2 What is still missing

Three gaps motivate this work. First, most systems, including DDI-GPT and DDIAgents, operate on drug pairs, whereas real polypharmacy commonly involves three or more medications. HODDI [8] explicitly targets high-order interactions, but is a benchmark with GNN and hypergraph baselines rather than an end-to-end multi-drug prediction-and-explanation system.

Second, locally hosted LLMs and chained multi-agent pipelines can be resource-intensive and add substantial inference latency. This project instead uses a single API-based LLM over a retrieved subgraph, prioritising simplicity and low latency while extending to multi-drug queries. Third, retrieval alone can expose only interactions already represented in the graph; it cannot readily

estimate calibrated probabilities for genuinely untested combinations. A trained graph model is therefore a future extension. Predictions also remain population-level and explanations are not necessarily clinician-ready.

1.3 Objective and Project Scope

1.3.1 Objective

Taking DDI-GPT as a baseline reference, this project develops an end-to-end clinical decision-support framework that predicts and explains adverse effects for regimens of three or more drugs. The framework establishes an API-driven Graph-RAG pipeline over the Decagon biomedical graph. Instead of a heavy locally hosted pairwise model, a lightweight API-based LLM orchestrator processes multi-drug cocktails simultaneously and synthesises multi-hop biological pathways into clinically interpretable rationales with minimal inference latency.

For a queried drug set, the system retrieves a local subgraph containing the drugs, their target proteins, protein–protein links among those targets, and known side-effect edges. It serialises this evidence into compact context, from which an API-based LLM predicts likely side effects and provides a natural-language explanation.

1.3.2 Dataset and suitability

The Decagon dataset from BioSNAP at Stanford is a standardised resource for polypharmacy side effects. It is provided as three connected CSV files:

- `bio-decagon-targets.csv`: human proteins targeted by approximately 645 drugs.
- `bio-decagon-ppi.csv`: physical interactions among approximately 19,000 human proteins.
- `bio-decagon-combo.csv`: observed side effects, across 964 types, for co-prescribed drug pairs.

In full, the graph contains 645 drugs, 19,085 proteins, 18,596 drug–protein edges, 715,612 protein–protein edges, and 4,651,131 drug–drug side-effect edges. Standard identifiers connect the three files without manual labelling or external scraping. The entities map directly to drug and protein nodes, and the relations map to typed drug–protein, protein–protein, and drug–drug side-effect edges. This structure makes a compact local subgraph available for Graph-RAG.

1.3.3 Deliverable and boundaries

The deliverable is a functional, low-latency API pipeline that accepts multi-drug queries, dynamically extracts a local subgraph, and returns qualitative side-effect predictions with mechanistic biological rationales. Training a graph neural network to predict novel unobserved adverse edges is outside the current phase. Continuous clinical variables—dose, pharmacokinetics, and renal/hepatic clearance—and empirical wet-lab validation are also future work.

1.4 Report Structure

Table 1.1: Organisation of the report

Chapter	Content
1 — Introduction	Problem and motivation, related work and gaps, objective and scope.
2 — Background and Related Work	Knowledge graphs, Graph-RAG, and LLMs for biomedical prediction.
3 — Data and Methodology	Decagon knowledge-graph construction and the Graph-RAG + API-LLM pipeline.
4 — Experiments and Results	Implementation, baselines, and metrics.
5 — Discussion	Interpretation, limitations, and practical implications.
6 — Conclusion and Future Work	Outcomes and directions for extension.

Background

2.1 The Knowledge Graph

The biomedical knowledge graph contains two node types—drugs and proteins—and three relation types (Table 2.1). The drug–drug relation is sourced from Decagon.

Table 2.1: Relation types in the knowledge graph

Relation	Description
Drug → Protein	Proteins a drug binds to or modulates. This is a property of an individual drug and does not depend on co-prescribed drugs.
Protein ↔ Protein	A fixed, drug-independent network of physical protein interactions, including signal transduction and shared pathways.
Drug ↔ Drug	Previously documented side effects for specific drug pairs, as recorded in Decagon.

Only the third relation is inherently pairwise: each polypharmacy edge connects exactly two drugs. Drug–target and protein–protein relations remain available for any number of drugs.

2.2 Standard Graph-RAG

Graph-RAG has two stages. During retrieval, the graph is traversed outward from entities named in a query to collect a relevant neighbourhood, preserving relational structure rather than relying only on textual similarity. During generation, that neighbourhood is serialised as text and supplied to an LLM, allowing the model to reason over explicit structure rather than memorised associations. Retrieval is not intrinsically limited to two entities because it can expand from any query set.

2.3 Problem Statement

Given a drug regimen of $k \geq 3$ drugs, the task is to predict potential side effects. Each prediction is annotated with an interpretive tier and the supporting path in the retrieved subgraph.

Methodology

The pipeline operates on the Decagon knowledge graph. Figure 3.1 illustrates the data structure: drug and protein nodes are connected by drug–protein, protein–protein, and pairwise drug–drug side-effect edges. Retrieval traverses the first two edge types for every drug in a queried regimen, while drug–drug edges supply known pairwise history.

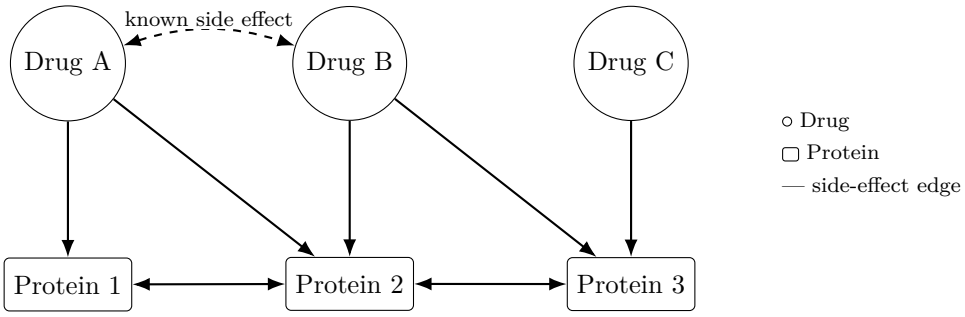


Figure 3.1: Example Decagon knowledge graph and the relation types used for retrieval.

The proposed pipeline (Figure 3.2) has three stages: retrieval of the queried drugs' neighbourhood, pruning of structurally noisy proteins, and generation of tiered, traceable predictions.

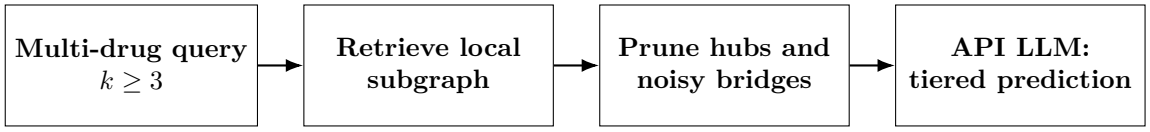


Figure 3.2: Overview of the proposed Graph-RAG pipeline for a regimen of $k \geq 3$ drugs.

3.1 Extending Retrieval from Two to $k \geq 3$ Drugs

Retrieval over drug–target and protein–protein relations is conducted independently for each drug, then combined; it therefore generalises to any k without modification (Table 3.1).

Table 3.1: Retrieval sequence for a regimen of $k \geq 3$ drugs

Step	Operation	Description
1	Look up each drug's targets	For every drug in the regimen, retrieve its target-protein set independently.
2	Combine target sets	Take the union of all targets to form one neighbourhood.
3	Find protein bridges	Find direct target-to-target protein edges or paths through one intermediate protein.
4	Find shared targets	Identify proteins targeted by three or more drugs by intersecting the target sets.
5	Attach pairwise history	For every drug pair, add documented side effects so the LLM knows existing evidence.
6	Serialise	Resolve identifiers to standard drug and gene names; convert drugs, targets, bridges, shared targets, and history to text for the LLM.

Bridges are restricted to at most two hops. Longer paths would introduce many indirectly related proteins and make the neighbourhood noisy; the restriction keeps each bridge biologically direct. Shared targets expose a signal unavailable to pairwise analysis: a protein targeted by three drugs becomes visible only after their neighbourhoods are combined. Proteins targeted by exactly two drugs remain in the neighbourhood and are covered by pairwise history. The assembled context prompts the LLM to reason from target perturbation to possible adverse outcomes before returning a final answer.

3.2 Pruning Before Passing to the LLM

Housekeeping proteins can interact with very many proteins and flood a neighbourhood with mechanism-irrelevant information. They are filtered before serialisation. A direct target of any regimen drug is always retained; other proteins are retained only if their global protein-network connectivity is below a chosen threshold. A protein above that threshold is treated as a generic hub and removed together with bridges that depend on it.

3.3 Output: Tiered, Traceable Predictions

The LLM output is organised into three tiers according to the evidence supporting a prediction (Table 3.2).

Table 3.2: Interpretive tiers of predicted side effects

Tier	Trigger	Clinical meaning
1: Known pairwise effect	A documented side effect exists for one drug pair in the regimen.	An independent effect of that pair, unrelated to the other drug(s).
2: Amplified effect	A protein is a shared target of three or more drugs, and a pairwise effect is documented for at least one pair among them.	A documented effect may be escalated because several drugs act on the same target.
3: Newly emergent effect	A bridge links targets of two different drugs and the predicted effect is not documented for that pair.	A potential adverse event with no pairwise precedent, arising only from the combined regimen.

The model is constrained to a fixed structured format (for example, JSON). For every predicted side effect it must state the assigned tier, a supporting subgraph path, and a plain-language explanation. Each cited path is automatically checked against the retrieved subgraph; a prediction with no valid path is discarded. Tier 2 and Tier 3 claims are thus traceable to retrieved structure and machine-readable for downstream use. Tier 3 predictions remain hypotheses requiring experimental or clinical validation.

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